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Fundamentally-based kinetic model for propene pyrolysis

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ABSTRACT

The primary objective of this work is to develop an improved fundamentally-based mechanism that describes the molecular weight growth kinetics observed during propene pyrolysis. Earlier attempts to describe the kinetics in terms of theoretically plausible reactions generally under-predicted the low temperature reactivity. To address this issue, propene pyrolysis experiments were performed at 575–875 °C with nominal residence times of ~2.4, 1.2 and 0.5 s at ~0.83 atm. These data were compared to a kinetic model that includes several reactions that involve allyl radicals. Specifically, electronic structure calculations at the CBS–QB3 level were performed for various allyl radical reactions, including addition, recombination, and hydrogen abstraction. This updated model is able to capture the observed fuel conversion, production of major products, and formation of molecular weight growth species. The sensitivity and rate of production analyses show that allyl reactions play important roles for both fuel conversion and product formation. In particular, allyl addition to propene leads to production of CH₃ and H. The H-atoms can add to propene to form CH₃ radicals, and both CH₃ and H can abstract from propene to regenerate allyl, completing the reaction chain. This model also successfully predicts the fuel conversion and major products for selected literature propene pyrolysis data.

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1. Introduction

Hydrocarbons, whether derived from petroleum or renewable sources, will be a primary fuel source for the foreseeable future. Thus, it is essential to be able to accurately predict their reactivity to optimize both the fuel production processes in the refinery as well as the multiple energy conversion processes that employ these fuels. A particular challenge in many processes is the formation of molecular weight growth (MWG) species. For example, the pyrolysis of hydrocarbon feeds in steam crackers leads to formation of MWG species that are large enough to precipitate from the gas phase and form deposits, limiting the run time between decoking operations [1,2]. Similarly, the formation of MWG deposit precursors upstream of reformers used to convert hydrocarbon gases to synthesis gas limits the time between reformer catalyst regeneration cycles [3]. The formation of deposit precursors in the anode channel of a solid oxide fuel cell is problematic for similar reasons [4,5]. In severe cases, deposit formation can reduce the efficiency and even lead to a catastrophic failure of the conversion process. Furthermore, MWG species contained in exhaust gas emissions create environmental and health problems [6–8]. For these reasons, more research is needed to better characterize the hydrocarbon pyrolysis reactions that are responsible for much of the undesired MWG chemistry.

The first steps in the thermal decomposition of saturated hydrocarbons are well understood. C–C bond fission reactions produce radicals that subsequently undergo β -scission reactions to generate smaller radicals and olefins. Hydrogen atoms, methyl and other alkyl radicals formed in these reactions will abstract hydrogen atoms from the fuel, speeding up its consumption. While these reactions are rather well characterized in terms of their rate expressions, the subsequent chemistry of the olefins that are produced by the β -scission reactions is less well known. Olefins form a large fraction of the initial products and their subsequent reactions have a substantial impact on the final product distribution. Thus, an accurate olefin pyrolysis reaction subset is essential to properly characterize both hydrocarbon pyrolysis and oxidation. In particular, MWG formation starts at a much lower fuel conversion during olefin pyrolysis, compared to paraffin pyrolysis, and higher concentrations of MWG species are generated [9]. As a result, olefin pyrolysis provides a better opportunity to explore the MWG kinetics.

Many studies have focused on characterizing MWG chemistry at the higher temperatures associated with combustion [10–17]. Under these conditions two important species are acetylene and propargyl radical [18–20]; however, the concentration of these species falls off rapidly with decreasing temperature and the species that are responsible for MWG chemistry at lower temperatures are different. Under conditions that are typical in refinery processes (500–1000 °C), the reactions of olefins and allylic radicals are much more important [21]. In olefin pyrolysis, the formation of resonantly stabilized allylic radicals is favored energetically due to the lower corresponding C–H

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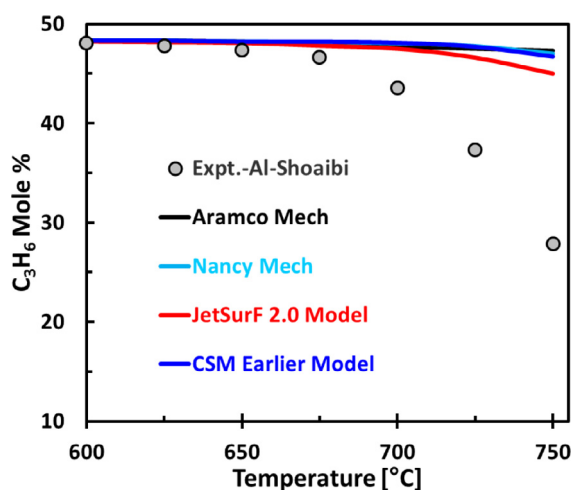


Fig. 1. Comparison between experiment and model predictions for the fuel conversion of propene (symbols—experiment from Al-Shoaibi [9] (EXP 1A), operating conditions: ~50% propene in N₂, τ = ~2.4 s, P = ~0.8 atm; lines—predictions of literature models).

and C–C bond dissociation energies in the parent olefin [22]. Because of their stability (when compared to alkyl radicals), allylic radicals can accumulate to relatively high concentrations. Thus, even though the rate coefficients for their various reactions are small, the rates of these reactions may be significant. Recently, we investigated the reactions of allylic radicals and identified several pathways that can lead to MWG species formation [21]. The investigated reactions include self/cross-recombination, addition to olefins, and abstraction from olefins. Recombination reactions lead to formation of a stabilized adduct that can potentially undergo hydrogen atom abstraction followed by unimolecular cyclization and β -scission reactions. Depending upon the conditions (e.g., higher pressures), these pathways can lead to the formation of stable MWG species. The addition of allylic radicals to olefins can also lead to MWG species formation. Again, cyclization of the adduct followed by β -scission is an important energy accessible route. While hydrogen abstractions of olefins by allylic radicals do not directly lead to the formation of MWG species, these pathways do compete with the addition reactions and can shift the relative concentrations of the various stabilized radicals.

The objective of this study is to characterize the MWG chemistry that occurs during olefin pyrolysis. This paper focuses on propene (C₃H₆) pyrolysis. (Subsequent work will examine the pyrolysis of butene isomers.) Propene is the simplest olefin with an allylic site; yet, despite several prior investigations [9, 23–29], questions remain about its low temperature kinetics. Earlier attempts to describe the kinetics in terms of theoretically plausible reactions generally under-predicted the low temperature reactivity. For example, the earlier kinetic model developed in our group was able to properly characterize alkane pyrolysis [9, 30,31], but significantly under-predicted the propene conversion [9] as shown in Fig. 1. The figure also shows that several other published kinetic models [32–34] also fail to capture the fuel conversion. These particular models were primarily developed to describe alkane/alkene oxidation, so these comparisons are under conditions that are well removed from those under which they were developed. Generally they perform well under higher temperature pyrolytic conditions rather than at the lower temperatures here. Nevertheless, this comparison does illustrate that significant uncertainties remain. It is essential to first understand fuel conversion in a system as simple as propene pyrolysis; only then it is possible to unravel the kinetics for the formation of MWG products.

Several investigations have reported the formation of MWG species during propene pyrolysis. Simon and Back [23,24] performed propene pyrolysis in a static reactor, over a temperature range 743–

873 K and a pressure range 200–600 Torr. They observed the early formation of C₅ and C₆ products in addition to the primary products, methane, ethene, and propane. The primary C₆ products were identified as 2-hexene, 1-methyl-cyclopentene, 3-methyl-cyclopentene, while the C₅ species was cyclopentadiene. They suggested that, following an initiation step, a simple scheme of addition, abstraction, and decomposition occurs. Barbe et al. [25] also identified several MWG species during propene pyrolysis at low extent of conversion; these include 3- and 4-methyl-cyclopentenenes, 1,5-diallyl, 1-hexene, methyl-cyclopentane, 4-methyl-1-pentene, and several other C₅ or C₆ products that were not identified due to their very low concentrations. Hidaka et al. [26] and Davis et al. [27] measured the production of benzene during propene pyrolysis as a function of temperature and residence time. More recently, Norinaga et al. [28,29] studied the pyrolysis of propene as well as acetylene and ethene at pressures of 2–15 kPa and temperatures of 1073–1373 K. They were able to identify multiple MWG species using on/off-line gas chromatography. The model developed in that study was extensively validated for hydrogen, small hydrocarbons, and aromatics. It provided good predictions for acetylene and ethene pyrolysis, but markedly over-predicted the propene pyrolysis at lower residence times and under-predicted it at longer residence times. Part of the reason for the discrepancy might be that only high-pressure rate constants were used in the kinetic model and hence the pressure dependence was not considered.

In this study, a fundamentally-based mechanism that describes the kinetics that occurs during propene pyrolysis is presented. This mechanism is validated against propene pyrolysis flow reactor experiments that were performed at 575–850 °C with nominal residence times of ~2.4, 1.2, and 0.5 s at ambient pressures. The relatively long residence times (compared to those of a typical steam cracker) were chosen to enhance the MWG kinetics. These conditions result in fuel conversions that range from virtually no reaction to ~85%. Multiple MWG species ($\geq$ C₅) are identified and quantified. Comparison of the model predictions with experimental data yielded very satisfactory results in terms of the temperature dependence of propene conversion, major products, and MWG species. Sensitivity and rate analyses were used to identify the important reactions. The results show that the allyl addition reaction to propene plays a critical role in forming MWG species. This reaction also leads to production of CH₃ and H. The H-atoms can add to propene to form CH₃ radicals; both of these radicals can abstract from propene to regenerate allyl, completing the reaction chain. The kinetic model is also used to predict previously published propene pyrolysis data that were obtained under very different operating conditions. Several published models are also used here to model the data collected in this work; we demonstrate that incorporating the allyl addition to propene into these models enables them to properly describe propene conversion.

2. Methods

2.1. Experimental description

Propene pyrolysis experiments were conducted using a continuous flow tubular reactor. Two sets of experiments are reported here. The first data set [9] (EXP 1A) was collected using a simpler version of the flow reactor apparatus, which has previously been described [30]. Several years later, the data set was reproduced (EXP 2A) and expanded upon (EXP 2B and 2C) using a more comprehensive detection scheme. Since the second set of data is more complete, the paper will primarily focus on this set. A diagram of the apparatus employed in the second set of experiments is provided in the supporting information (SI), Fig. S1.

A known flow of ~50% propene (Matheson TriGas Co., 99.9%) in N₂ (General Air, 99.998%) was passed through a 6 mm-ID tubular quartz reactor that was housed in an electric furnace. Some impurities in the propene, including ~0.05% propane and ~0.005% 1,5-hexadiene,

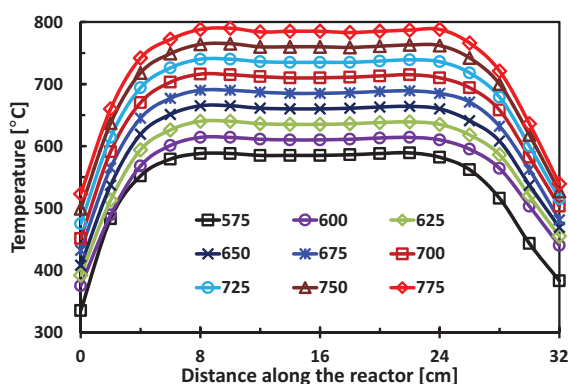


Fig. 2. Temperature profiles measured along the centerline of the flow reactor.

were measured and these were accounted for in the modeling efforts. Experiments were conducted at a (high-altitude) ambient pressure of ~ 0.82 atm. The temperature profile of the reactor was measured axially along the length of the reactor using a K-type (Cr/Ni) thermocouple. As shown in Fig. 2, the temperature was approximately constant in the central region of the reactor. The temperature in this region was varied from 575 to 850 °C; the reproducibility of the measurement is ± 2 °C. At the entrance and exit of the reactor the temperature gradient is steep, and the measurement is very sensitive to the exact axial position of the thermocouple. An estimated maximum error of ± 15 °C occurs at the first and last position of the profile. However, the temperatures at the edges of the reactor are sufficiently low such that chemistry is not expected to occur, and toward the center of the reactor the magnitude of this error decreases rapidly. The total flow rate was set to 30 (EXP 1A and 2A), 60 (EXP 2B), or 150 (EXP 2C) SCCM, which corresponds to nominal residence times of ~ 2.4 , 1.2, and 0.5 s, respectively, for the center constant temperature region of the reactor. The actual values are dependent upon the temperature profile, total flow, and extent of reaction (i.e., total number of moles in the reactor at a given distance) and are determined more precisely by model calculations that account for each one of these variables. Table S1 in the SI lists the temperature-residence time profiles for selected cases.

Following the reactor, the gas stream was analyzed using gas-chromatography (GC). In the first set of experiments (EXP 1A) a Hewlett Packard 5890 Series II Plus GC was used. Separation of the gas mixture was achieved via two analysis schemes. In both cases, the initial oven temperature was held at 40 °C for 5 min and then ramped at 20 °C/min to 220 °C. Permanent gases and light hydrocarbons ($\leq C_2$) were separated using tandem Supelco 6 ft $\times$ 1/8 in. stainless steel packed Porpack R and 15 ft $\times$ 1/8 in. stainless steel packed Carboxen 1000 columns with argon carrier gas and a thermal conductivity detector (TCD). In the second scheme, light and heavy hydrocarbons were separated using a J&W Fisher SPB-1 60 m $\times$ 0.53 mm-ID 5 μ m fused silica film capillary column with helium as the carrier gas and a flame ionization detector (FID). In the second set of experiments (EXP 2A, 2B, and 2C), the separation scheme for the permanent gases and light hydrocarbon remained the same, except that a FID was coupled to the TCD. A Thermo Scientific Finnigan Trace GC Ultra, with the same column, carrier gas, and oven ramp profile as in the first set of experiments, was used to separate the hydrocarbons. The column effluent was analyzed by a parallel FID and a mass spectrometer (MS) detector. The use of the MS detector in the second set of experiments allowed for a more definitive product identification. One limitation to the analytical method is that acetylene and ethene, 1-butene and isobutene, 1,3-methylcyclopentadiene and 2,4-methylcyclopentadiene, as well as 1,3-cyclohexadiene and 1,4-cyclohexadiene are not completely separated from one another and, in these cases, the sum of the two species is reported. Since the two

species that co-elute have similar response factors, the sum of the mole fractions and the carbon (C) and hydrogen (H) balances was only minimally impacted.

Calibration of the GC response factors was performed using either gaseous or liquid species. In the first set of experiments, standard gas mixtures were prepared in-house. The response factors of species that were not calibrated directly were estimated based on a correlation of response factors to the number of carbon atoms in the product for FID measurements. In the second set of experiments, commercial mixtures were used in addition to in-house mixtures. The commercial gas mixture (Matheson Trigas) contains nitrogen, methane, ethene, ethane, propene, propane, isobutene, and *n*-butane. The calibration of the liquid C_5 – C_9 species was performed by dissolving the species in benzene or toluene, then pumping this mixture through a vaporizer and into the reactor. A standard chemical solution (Sigma-Aldrich) containing seven C_6 – C_9 species (benzene, toluene, ethyl-benzene, *o*-xylene, 1,2,4-trimethylbenzene, 2-hexanone, and 2,2,4-trimethylpentane) was employed. The accuracy of the gas calibration is estimated to be $\sim 2\%$ relative, while that of the liquid calibration was estimated to be $\sim 5\%$ relative. Again, the response factors of species that were not calibrated directly (mostly larger hydrocarbons) were estimated, although in this set of experiments the estimations were based on a larger set of calibrations. Since the number of moles of N_2 gas does not change during the reaction, the change in this mole fraction allows for the conversion of the product mole fractions to their molar concentrations.

2.2. Kinetic modeling

The initial starting point for mechanism development was that used in our earlier work describing the MWG kinetics during ethane pyrolysis at high conversion [31]. This mechanism was expanded to describe the pyrolysis of propene and butene isomers; the results of the butene isomer study will be presented in forthcoming manuscripts. The mechanism contains ~ 580 species and ~ 9200 reversible reactions. While it is not feasible to discuss each reaction contained within the mechanism, they can be broken down into six main classes: (1) dissociation/recombination, (2) hydrogen atom abstraction, (3) addition/ β -scission, (4) isomerization, (5) disproportionation, and (6) other rearrangements such as Diels–Alder/retro-Diels–Alder reactions. Some specific reactions will be discussed below.

When reliable experimental rate constant data are available, these values are implemented in the mechanism. In other cases, rate constants are determined using transition state theory based on the results of electronic structure calculations, generally performed at the CBS–QB3 level of theory [35]. In many cases, this approach is used to develop rate estimation rules that can be applied to larger species that are not calculated directly. Detailed discussions of this approach have previously been reported [36,37]. Roughly one-third of the reactions in the mechanism are pressure dependent. The rate coefficients for these reactions are evaluated as a function of temperature and pressure using a three frequency Quantum–Rice–Ramsperger–Kassel analysis with a modified strong collision approximation (QRRK/MSK) [38]. The reliability of this approach has previously been discussed by several studies [39–42]. It is chosen because its approximate nature allows for one to efficiently generate results for larger species. This feature is essential for the development of large mechanisms. Thermodynamic reversibility is used to determine reverse rate constants. Thermodynamic parameters are either estimated from group additivity using THERM [43] or obtained from electronic structure calculations. Since bond additivity corrections were not applied to the results of the electronic structure calculations, there are small differences in the absolute heats of formations relative to the group additivity estimates. As a result, all reactions where the thermo was based on the electronic structure results were written irreversibly (in both

directions, using the CBS–QB3 derived thermo to determine the reverse rate constants) so as to keep the thermo internally consistent. The mechanism was used for this study without any attempts to improve the predictions by adjusting rate expressions. The files containing the complete kinetic model and the associated thermodynamic data are provided in the SI.

The simulations were performed using the ChemKin Pro suite of programs [44]. The input parameters consist of the kinetic mechanism and corresponding thermodynamic properties as well as the measured temperature profiles, flow rates, pressure, and reactor dimensions. The quartz reactor used in this study is treated as a plug flow reactor assuming that there are no wall effects. The plug flow assumption has previously been verified by comparing these results to those obtained using a parabolic flow model [30]. This program was used to generate concentration-time profiles, rates of production plots, and sensitivity analysis. In many instances, the mechanism was run in an irreversible format (i.e., the forward and reverse reactions are both explicitly specified). Note that in either format, reversible or irreversible, the reverse rate constants are calculated from the thermodynamic data base, using the group-additivity based or electronic structure derived thermochemistry as appropriate. Both formats provide identical concentration-time profiles. But, in a reversible format a sensitivity analysis will not identify partially equilibrated reactions. This is because an increase in the forward rate constant will be offset by the simultaneous change in the reverse rate constant. However, in an irreversible format the two reactions are now decoupled and the results can be used to examine the impact of the equilibrium constant assignment on the predictions.

In several places throughout the text, tables, and figures, we employ a chemical notation that omits the hydrogen atoms. For C_3 and larger species, the radical site is indicated by “•”, a double bond by “=”, a triple bond by “≡”, and a cyclic species by “cy”. The units employed are kcal, s, and mol.

3. Results

The results for all four sets of experiments are provided in the SI (Tables S2–S5). Each data point is an average of at least three measurements. It is important to note that the change in the measured mole fractions (with temperature or residence time) is due to both reaction and dilution since the total number of moles changes. The impact of dilution can be assessed from the change in the N_2 mole fraction, since the number of moles of N_2 remains constant. In the first data set (EXP 1A), only C_4 and smaller species were identified. For larger species, only the number of carbon atoms is given. In the second set of experiments (EXP 2A, 2B, and 2C), most of the products were identified. For the few species that remain unidentified, the molecular formula is known. For each experiment, the measured mole fractions sum to one, within the expected uncertainty of the experiments (see the third column of Fig. 3). This suggests that we are correctly quantifying the gas mixture that is contained in the GC sample loop. The C and H atom balances are also approximately unity. This means that essentially all of the products that are produced in the reactor, even the heavy ones, are making it to the GC sample loop. These results provide confidence that we are properly accounting for the vast majority of the reaction products.

The propene decomposition and sequential formation of the MWG species (per C number) are shown in Fig. 3 as a function of temperature. The two data sets collected at ~ 2.4 s (EXP 1A and 2A in the bottom row) are in good agreement with one another. The data sets collected at ~ 1.2 s (EXP 2B) and ~ 0.5 s (EXP 2C) are shown in the middle and top rows, respectively. The formation of C_4 – C_6 species occurs at very low levels of conversion. C_4 species are formed first, followed by C_5 and C_6 . The effect of temperature on formation of C_6 species indicates that its formation pathway does not necessarily proceed via the C_4 and C_5 species. The C_4 and C_5 profiles turn over at higher tem-

perature, as the formation of C_6 and C_7 continues to increase. The C_7 – C_{12} species do not start forming until the temperature is above 700 °C or higher. The mole fractions of the C_7 , C_9 , C_{11} and C_{12} species begin to decrease at higher temperatures, while the mole fractions of the C_8 and C_{10} continue to increase.

Fig. 4 shows the mole fraction-temperature profiles of propene and multiple reaction products, with the plots ordered in terms of the mole fraction of products formed at the three residence times. The data from EXP 1A and 2A are plotted together, showing good agreement. The major products are methane (CH_4), ethene (C_2H_4), and hydrogen (H_2). Other light species that are formed in lesser amounts are propyne ($C\equiv CC$), ethane (C_2H_6), allene ($C=C=C$), and propane (CCC). The C_4 species measured include 1,3-butadiene ($C=CC=C$), 1-butene ($C=CCC$), 2-butene ($CC=CC$), and *n*-butane ($CCCC$). The C_5 species include cyclopentadiene (CY13PD), 1,4-pentadiene ($C=CCC=C$), 2-methyl-1,3-butadiene ($C=CC_2=C$), 1,3-pentadiene ($C=CC=CC$), cyclopentene (cy C_5H_8), and 1-pentene ($C=CCCC$). Benzene (C_6H_6) is the dominant C_6 species. Other C_6 products are the 1,3- and 1,4-methyl-cyclopentadiene isomers (ME-CY13PD and ME-CY24PD), the 3-methyl-cyclopentene and 4-methyl-cyclopentene isomers (Et3-CYPENE and Et4-CPENE), the 1,3- and 1,4-cyclohexadiene isomers (CHD13 and CHD14), 1,5-hexadiene ($C=CCCC=C$), and cyclohexene (C_6H_{10}). Toluene (cy $C_6H_5CH_3$), naphthalene (a2), styrene (cy $C_6H_5C=C$), indene (C_9H_8), and ethyl-benzene (cy $C_6H_5C_2H_5$) are the primary C_7 to C_{10} species. Small amounts of 1,3,5-cycloheptatriene (C_7H_8), xylene isomers (C_8H_{10}), ethenyl-cyclohexene (C_8H_{12}), indane (C_9H_{10}), propenyl-benzene (C_9H_{10}), methyl-indene ($C_{10}H_{10}$), 1,4-dihydronaphthalene ($C_{10}H_{10}$), methyl-naphthalene ($C_{11}H_{10}$), biphenyl ($C_{12}H_{10}$), and acenaphthalene ($C_{12}H_8$) were also measured, but are not shown in Fig. 4 (see Tables S2–S5 in the SI).

The model predictions are also provided in Fig. 4. The model provides excellent predictions for the fuel conversion (panel a) and for the formation of the three major products, methane (b), ethene (c), and hydrogen (d) over the temperature and residence time ranges studied. The predictions are also quite good in capturing the profiles for the MWG species. The good descriptions for benzene, cyclopentadiene, toluene, naphthalene, styrene, and indene are particularly encouraging. The profiles for the intermediate MWG species all show strong temperature dependence; the observed mole fractions reach maxima and then decrease at higher temperatures. In most cases, the model is able to predict both the magnitude and observed temperature and residence time dependencies reasonably well.

4. Discussion

The favorable comparisons of model predictions to the extensive data sets suggest that the kinetic model captures the essential underlying kinetics. Thus, we are now in a position to use the model to explore the primary pathways that account for the fuel decomposition and for the formation of MWG species. The analysis presented here will primarily focus on EXP 2A at 775 °C. Under these conditions, the fuel conversion is high ($\sim 60\%$) and substantial amounts of MWG species are formed. Most of discussion will focus on the chemistry that occurs in the constant temperature region of the reactor. However, it is important to keep in mind that initiation chemistry occurs near the front of the reactor where the temperature is much lower (see the 775 °C temperature profile in Fig. 2). At the end of the reactor the chemistry is quenched when the temperature decreases.

4.1. Kinetics of fuel conversion

Several prior studies [26, 45–48] have investigated the thermal decomposition of propene under conditions that allow for the initiation chemistry to be explored (i.e., more dilute fuel concentrations, lower fuel conversion, and/or shorter residence times). Recently,

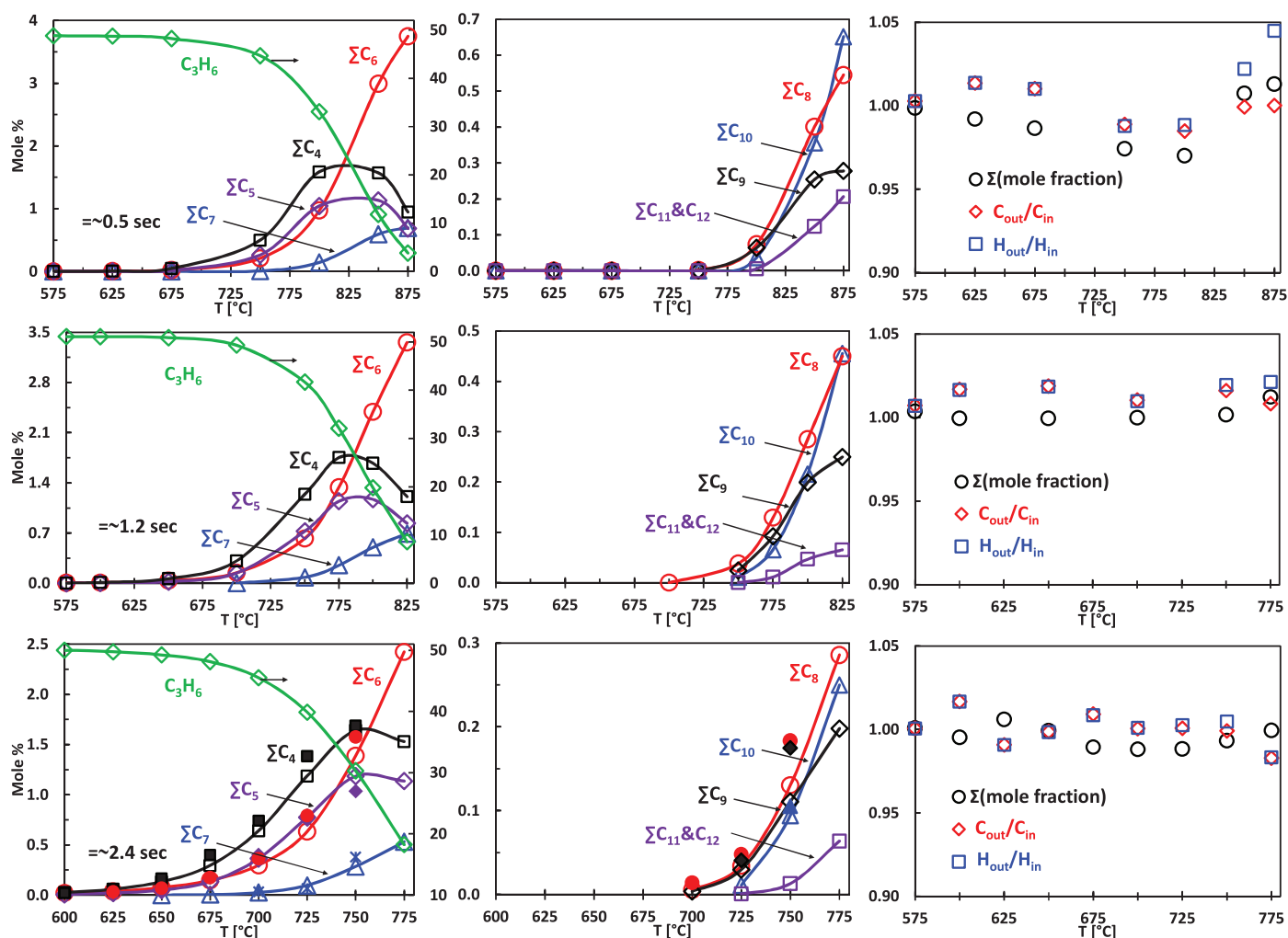


Fig. 3. Observed formation of MWG species (left and middle panel) with temperature and measured mole fraction sum and mass balances (right panel) for propene pyrolysis (operating conditions: initial ~50% fuel diluted in nitrogen, $P = \sim 0.8$ atm, row 1: $\tau = \sim 0.5$ s; row 2: $\tau = \sim 1.2$ s; row 3: $\tau = \sim 2.4$ s, filled symbols—Al Shoaibi [9] (EXP 1A), open symbols with lines—this work (EXP 2A, 2B, and 2C). Note that the lines are used to guide the eye; the mole fraction of propene is displayed on the right-hand side y-axis in the plots in the first column.

Ye et al. [49] conducted theoretical calculations for a number of reactions on the C_3H_6 potential energy surface (PES) using the CCSD(T)/cc-pVTZ method. Hung et al. [50] studied the unimolecular decomposition pathways using CBS-QB3 calculations coupled with RRKM/ME calculations. The results were compared to experimentally measured time-dependent H-atom profiles that were obtained behind reflective shock waves at 1450–1710 K at ~ 2 atm. Rate constants were determined for four pathways (R1)–(R4), and reaction (R1) was confirmed to be the dominant route under these operating conditions. The kinetic mechanism employed here contains these four decomposition channels. In addition, we include the two bimolecular channels (reverse disproportionation reactions, (R5) and (R6) recommended by Tsang [51].



The rate constants used in our model for the unimolecular reactions of propene (R1)–(R4) are consistent with those reported by Ye et al. [49] and Hung et al. [50]; the comparison is provided in the SI (Fig. S2). The reverse disproportionation reactions (R5) and (R6) are second order in propene; thus the relative importance of these reactions to the unimolecular reactions will depend upon the propene concentration. To facilitate the comparison of (R5) and (R6) to (R1)–(R4), we calculated pseudo-first order rate constants. The comparisons are shown in Fig. 5 for initial propene mole fractions of 50% (used in this work) and 0.1%. Under the temperatures and propene concentrations employed in this work, the reverse disproportionation reactions are generally faster than the unimolecular channels. The rate constants for the unimolecular channels only become comparable when the temperature increases to ~ 850 °C, near the upper limit of the temperature range of our experiments. In contrast, at lower concentrations of propene (e.g., 0.1%), the unimolecular decomposition pathways (R1) and (R2) become much more important.

A sensitivity analysis (with all reactions written irreversibly) was performed at 775 °C (Table S6 in the SI). This sensitivity analysis characterizes the impact of specific reactions in terms of changes in the propene concentration at the end of the reactor. The results indicate that only a handful of reactions are responsible for the decomposition of propene. In order, the four most important reactions are:

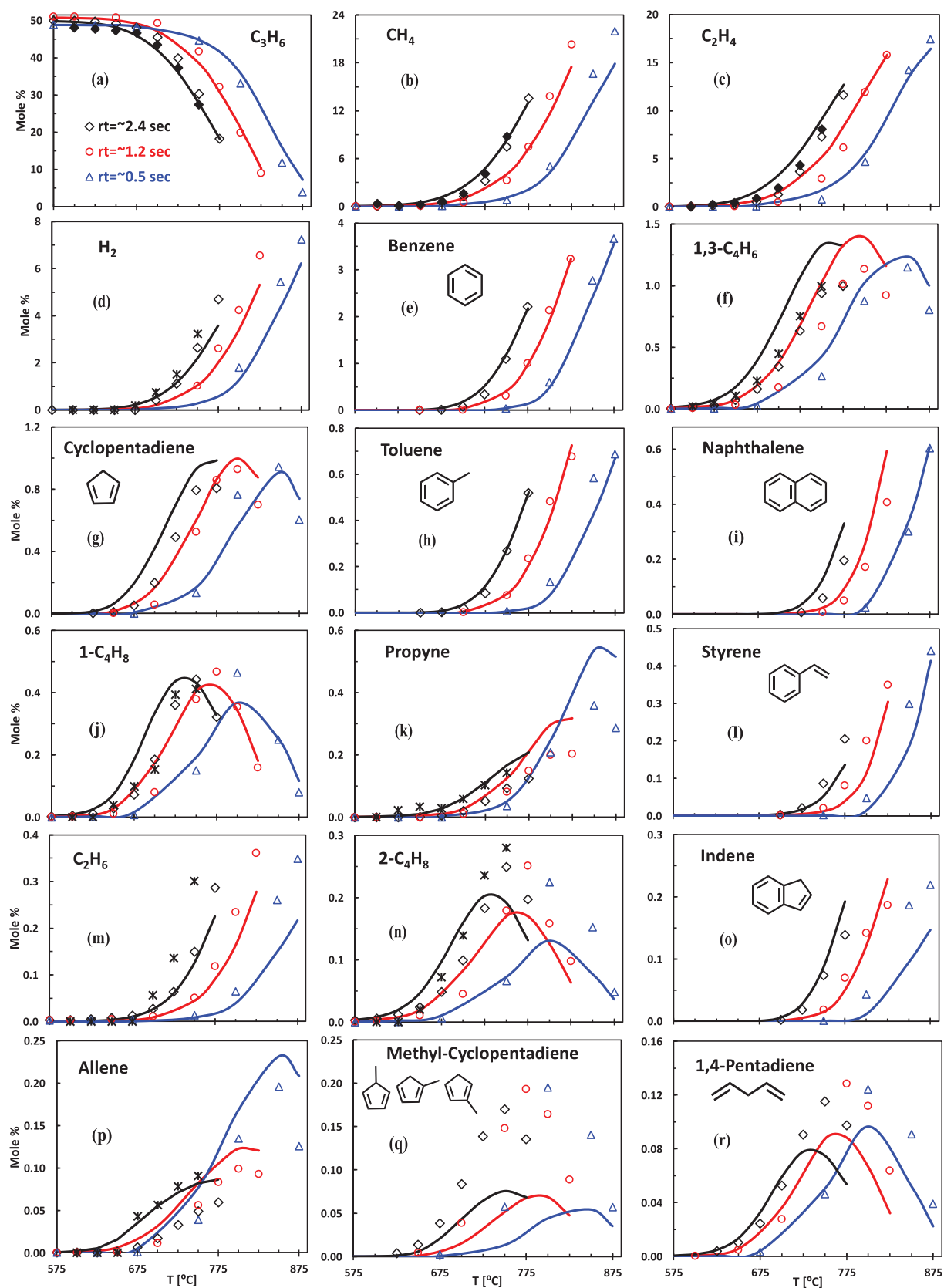


Fig. 4. Comparison of model predictions to products distribution in propene pyrolysis data. Symbols—experiment (filled symbols—Al Shoaibi [9] (EXP 1A). Open symbols—this work (EXP 2A, 2B, and 2C)); lines—modeling. Operating conditions: initial ~50% fuel diluted in nitrogen, $P \sim 0.8$ atm, $T = 575\text{--}875$ °C, $\tau \sim 0.5, \sim 1.2$, and ~ 2.4 s.

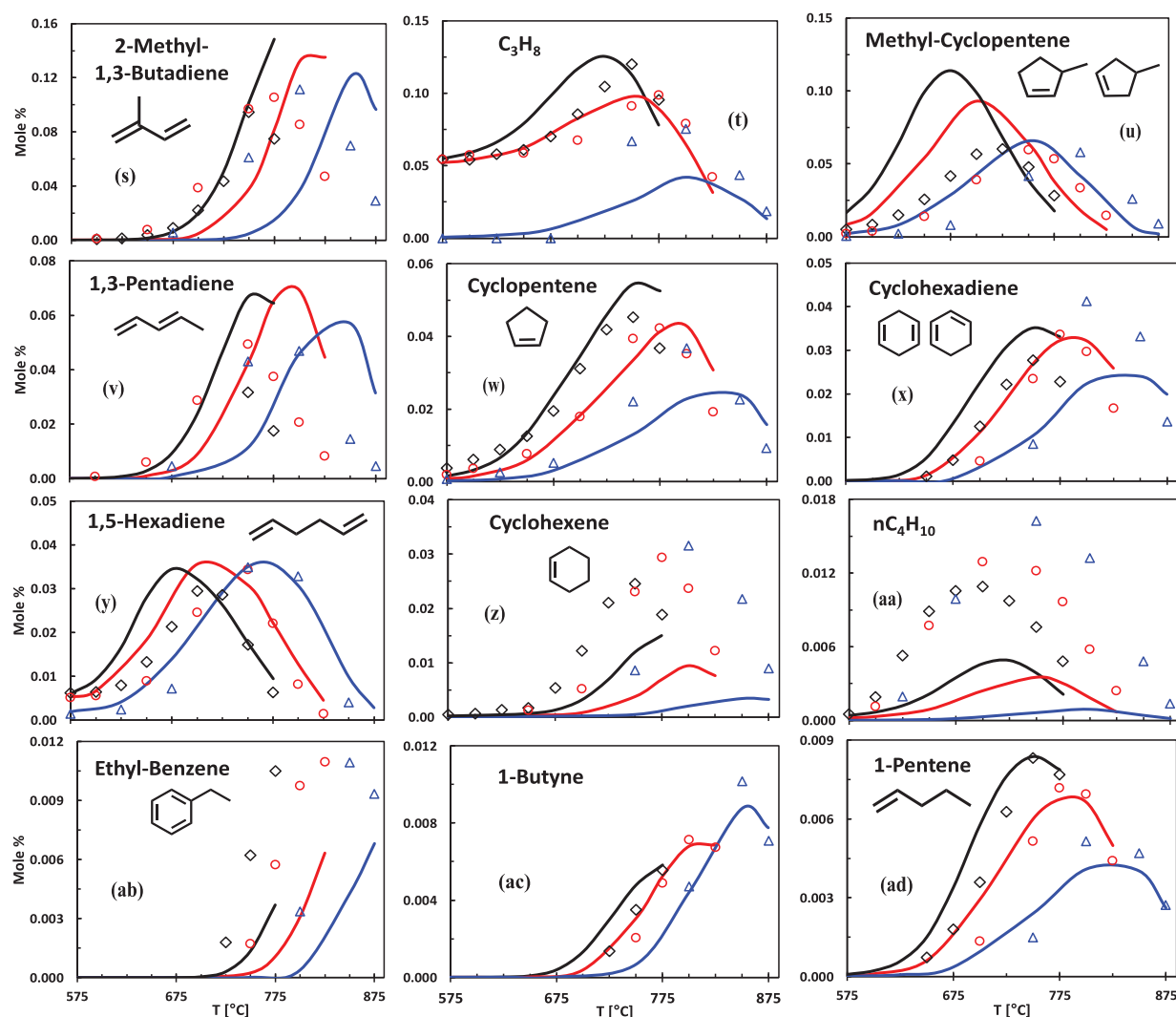


Fig. 4. Continued

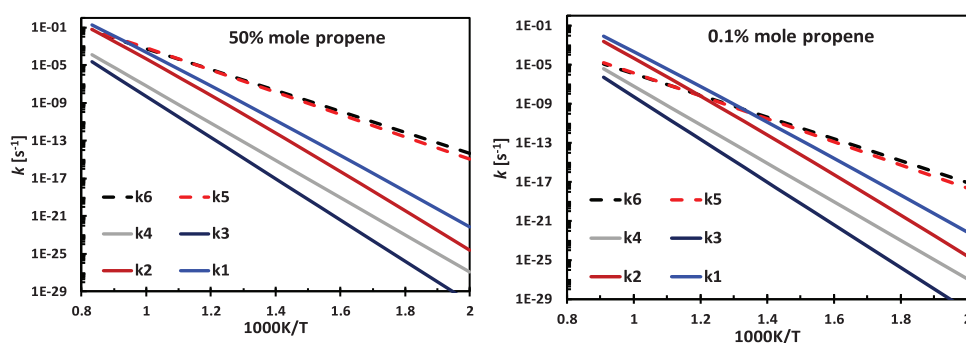
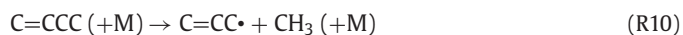
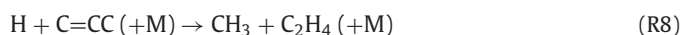


Fig. 5. Comparison of the reaction rate constants for the unimolecular and the pseudo-first order rate constants for the bimolecular channels under different fuel concentrations.



Reactions (R7)–(R9) directly remove propene, whereas (R10) and (R9) produce allyl radical that then can react with propene via (R7). Re-

action (R10) is nearly partially equilibrated and therefore the results are more sensitive to the equilibrium constant assignment than to the rate constants. (Here the reverse reaction has a similar magnitude sensitivity coefficient but with the opposite sign.) Interestingly, none of these reactions is related to the initiation process. This indicates that, under our particular conditions, the fuel decay is not sensitive to the rate parameter assignments for reactions (R1)–(R6). This implies that, once the first radicals are generated, the above reactions dominate propene conversion. Fig. 6(a) shows the rate of production of propene along the reactor. In addition to the above reactions, three

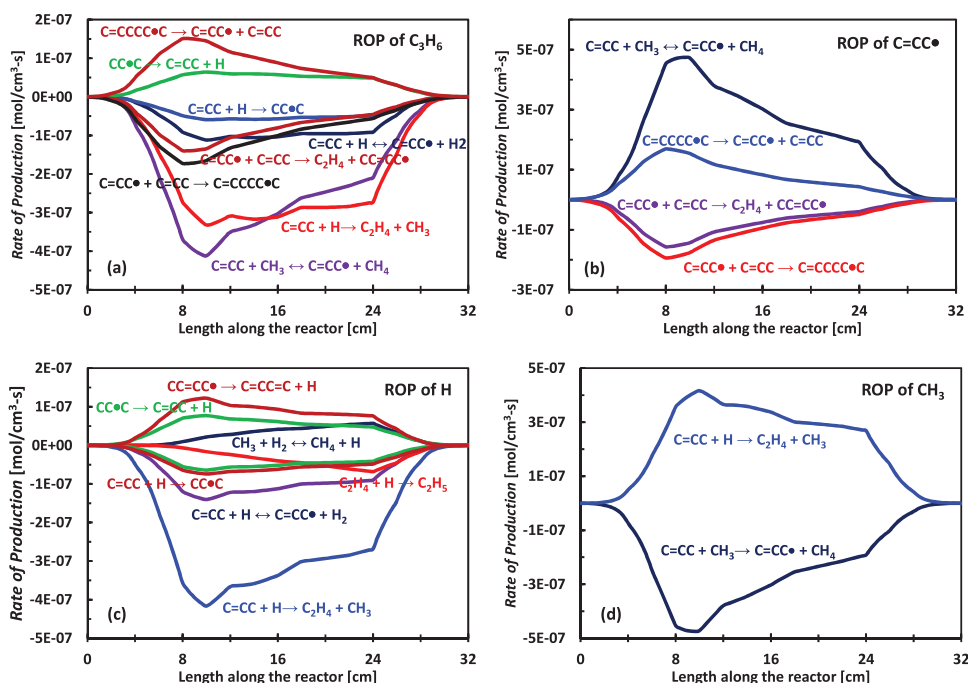
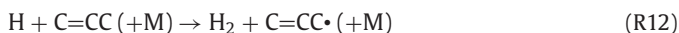


Fig. 6. Rate of production of propene (a), allyl (b), H atom (c), and methyl (d) along the reactor with residence time of ~ 2.4 s at 775 °C. (Most H atoms in the species names are omitted for clarity.)

other reactions (R11)–(R13) occur. The impact of (R11) and (R13) on the propene conversion is small since the reverse reactions also occur.



The reactions of H, CH_3 , and allyl radical play an important role in the decomposition of propene. The rates of production for these three radicals are shown in Fig. 6(b–d). These results show that propene decomposes via the following auto-catalytic cycle: Allyl radical is primarily formed by H-atom abstraction from propene by CH_3 (R9) and H-atoms (R12). Allyl addition to propene primarily leads to the formation of ethene + methyl-allyl radical (R7). The subsequent β -scission reaction of methyl-allyl radical forms H-atoms and 1,3-butadiene (R14).



H-atom addition to propene leads to the formation of CH_3 + ethene (R8). CH_3 and H then abstract from propene to form allyl radicals. Because allyl is so stable, its conversion to more active radicals is slow. Note that the rate of formation of allyl is greater than its rate of depletion. As a result, the thermal decomposition of propene starts at higher temperatures than the decomposition of saturated fuels, such as propane.

The rate constant parameters for these reactions are either from well-documented studies, or they are derived from high-level electronic structure calculations. The high-pressure limit rate constant of the reverse reaction of (R10) is taken from the Tsang recommendation [51]. Reactions (R8) and (R13) are part of the C_3H_7 PES, while (R7) and (R11) are part of the C_6H_{11} PES. These two surfaces were evaluated using CBS–QB3 calculations. The results from the C_3H_7 PES have previously been discussed [31]. The C_6H_{11} PES is shown in Fig. 7. In addition to (R7) (shown as exit channel (IV)) and (R11) (shown as

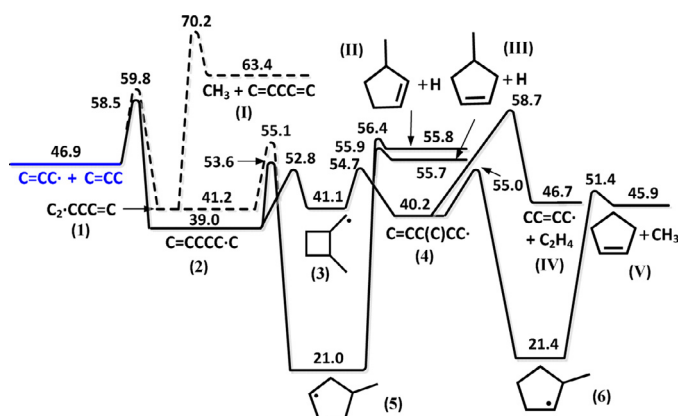


Fig. 7. Simplified C_6H_{11} PES calculated at the CBS–QB3 level of theory, enthalpies in kcal/mol at 298 K.

well (2)), there are other low-energy channels which can be accessed. A detailed discussion of this PES will be given in the next section. The two hydrogen abstraction reactions, (R9) and (R12), are also based on CBS–QB3 calculations. Fig. S3 in the SI shows that the rate constants for (R8), (R9), and (R12) used in this study are consistent with prior reported values.

As mentioned above, our earlier kinetic model [9,30,31] failed to capture thermal conversion of propene (cf. Fig. 1). This issue was primarily resolved by adding the addition reactions of allyl to propene that occur on the C_6H_{11} PES (Fig. 7) to the mechanism. As discussed earlier, several other mechanisms in the literature are also unable to capture thermal conversion of propene [32–34]. These mechanisms also do not consider the addition reaction of allyl to propene. Thus, the predictions from these models might also be improved by incorporation of the C_6H_{11} PES reactions. To test this hypothesis, this reaction subset was added to the JetSurF 2.0 model [33] and the Aramco mechanism [34]. As shown in Fig. 8, the predictions for the fuel conversion of propene are significantly improved with this update. Note that the problem of an incomplete description of propene initiation

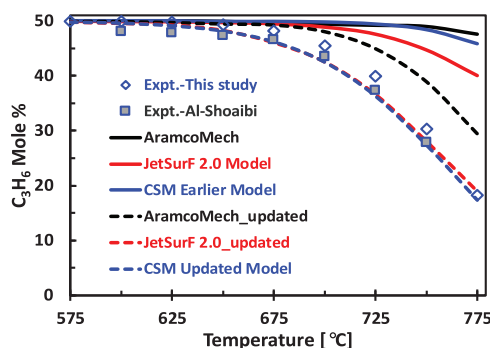


Fig. 8. Comparison between experiment and prediction of several known literature models for the fuel conversion of propene (open symbols—experiment by this study (EXP 2A); filled symbols—experiment from Al-Shoaibi [9] (EXP 1A); solid lines—the original literature models; dashed lines—the selected models updated by the reactions of C_6H_{11} PES).

kinetics under low temperature conditions with high fuel concentration and conversion may be not limited to the models considered here and the comparisons should not be taken as a criticism of these models, especially since they were primarily developed to describe oxidation and/or combustion kinetics. Rather, the comparison is intended to show the importance of the addition reaction of allyl to propene. Depending upon the conditions (i.e., temperature, pressure, residence time, equivalence ratio, etc.), the incorporation of this reaction sequence into these models might potentially impact the predictions under certain pyrolysis and oxidation conditions. In general, it is important to properly characterize the pyrolysis chemistry in oxidation/combustion models since it can set the stage for the oxidation kinetics.

4.2. Formation of light species

The three major products (hydrogen, methane, and ethene) are formed primarily by reactions (R12), (R9), and (R8), respectively. The formation of alkane species including ethane, propane, and *n*-butane are primarily from the H-abstraction of propene by ethyl, isopropyl, and 2-butyl, respectively. The PES forming ethyl, isopropyl, and 2-butyl have been described during the analysis of ethane pyrolysis [31]. Allene ($C=C=C$) is primarily formed by two disproportionation reactions [51]:



The dissociation of allyl provides a minor contribution to allene formation. Propyne ($C\equiv C$) is primarily formed from the isomerization of allene, via a cyclic- C_3H_4 intermediate [52].

4.3. Kinetics of molecular weight growth

An important kinetic feature for olefin pyrolysis is that relatively high concentrations of resonantly-stabilized free radicals (RSFRs) are generated. Fig. 9 shows the predicted mole fractions of the major radicals produced during propene pyrolysis. Most of the resonantly stabilized radicals are formed from an approximately thermoneutral H-atom abstraction reaction from the parent olefin by allyl radical. This type of abstraction reaction is important because it has the potential to shift the MWG product distribution [21]. Note that the mole fraction of allyl and cyclopentadienyl radicals are around 2 orders of magnitude higher than that of methyl, and 4–5 orders of magnitude higher than ethyl and H-atom. This difference may be attributed to two factors: (1) The rate of production of these species is higher

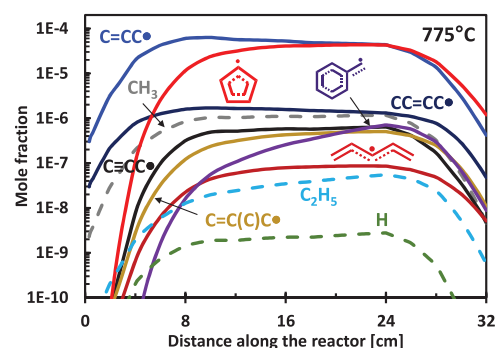


Fig. 9. Predicted mole fractions of dominant radicals in propene pyrolysis with residence time of ~ 2.4 s at 775 °C. (Most H atoms in the species names are omitted for clarity.)

due to the weaker C–C and C–H bonds in the parent olefin, and (2) the rate constants for the subsequent reactions are small. However, the higher concentration of these radicals may mean that the reactions may still be important. Recombination and addition reactions involving RSFRs are thought to be critical for the formation of MWG species [21].

There are several dominant resonant radical addition reactions that lead to MWG formation. As mentioned above, the thermal decomposition of propene is sensitive to (R7) and (R11), which are part of the C_6H_{11} PES (Fig. 7). On this surface, (R7) corresponds to the formation of bimolecular products (IV) where (R11) to the formation of (2). Allyl addition can occur on either side of the double bond resulting in the formation of 4-methyl-1-pentene-5-yl (1) and 1-hexene-5-yl (2). The barriers for addition are ~ 12 – 13 kcal/mol, about 5 kcal/mol higher than a typical addition reaction of an alkyl radical where there is no loss of resonance energy. The initially-formed adduct is much less stable than an analogous alkyl addition adduct. Once formed, the two linear adducts can cyclize via barriers that are lower in energy than those leading back to reactants. Endo-cyclization of both adducts leads to the formation of 3-methylcyclopentyl (5) and its subsequent β -scission leads to two methyl-cyclopentene isomers plus H (II and III). The 1-hexene-5-yl (2) adduct can also undergo exo-cyclization to form (3). This pathway has a slightly lower barrier than that for endo-cyclization; it also has a higher pre-exponential factor since one less hindered rotor is lost during cyclization. Due to the shallow well the cyclic radical (3) ring opens to form 4-methyl-1-pentene-5-yl (4), which can produce 1-methyl-allyl + ethene (IV) or re-cyclize to form 2-methylcyclopentyl radical (6) before undergoing a β -scission reaction to form methyl + cyclopentadiene (V). The key feature of this PES is that the cyclization reactions of the linear radicals (1), (2) and (4) and the subsequent β -scission reactions of the cyclic alkyl radicals (5) and (6) are all lower in energy than the entrance channel.

A QRRK/MSK analysis was used to obtain apparent rate constants at various pressures. Fig. 10 shows the predictions at 1 atm. For both addition pathways, the formation of the collisionally-stabilized adducts (1) and (2) dominates at lower temperatures. As the temperature increases, a significant fraction of the initially formed adduct dissociates back to the reactants due to the relatively shallow well, causing the overall rate constant k_{tot} to deviate from k_{∞} . Other channels also begin to become important. Interior addition leads to the formation of $CH_3 + 1,4$ -pentadiene (I) followed by H + methyl-cyclopentene isomers (II and III) and methyl cyclopentyl (5). Terminal addition leads to the formation of methyl-allyl + ethene (IV), followed by methyl cyclopentyl (5) and H + methyl-cyclopentene isomers (II and III). These results reveal that allyl is converted to more reactive radicals, H and CH_3 , either directly or indirectly. An important indirect path results from a subsequent β -scission reac-

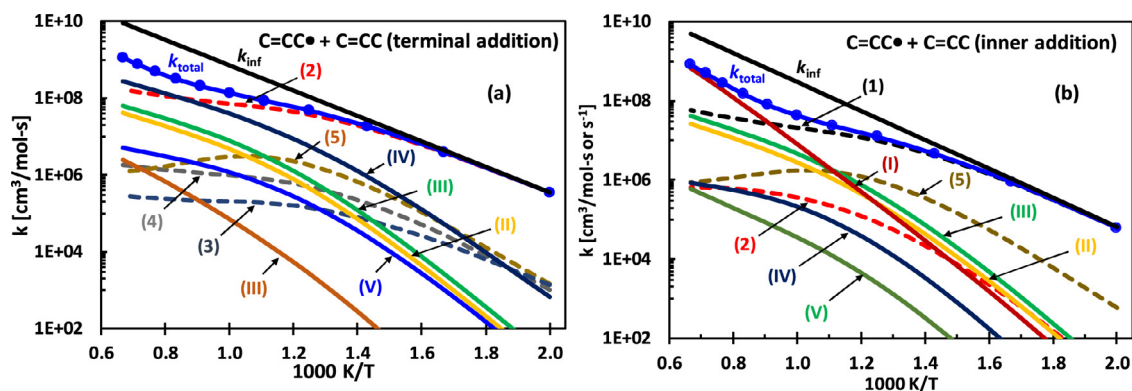


Fig. 10. Predicted apparent rate constants for the addition of allyl to propene at 1 atm in N₂ (left: terminal addition; right: nonterminal addition. Cf. Fig. 7 for the structures).

tion of methyl-allyl to form H + 1,3-butadiene (R14). Several unsaturated MWG species are also formed. These species (along with ethene) can undergo subsequent radical addition reactions. They also can form larger-sized resonantly stabilized radicals via H-atom abstraction, which may add to olefins, leading to the formation of MWG species. For example, the methyl-cyclopentene isomers and cyclopentene, can form large-sized resonantly stabilized radicals via subsequent hydrogen abstraction, leading to the formation of methyl-cyclopentadiene isomers and cyclopentadiene, which ultimately form aromatic species (e.g., benzene, toluene) (Scheme 1). Since the allyl radicals start forming once propene begin to react, the formation of these C₅ and C₆ species are expected to be formed even when the fuel conversion is very low. Dente et al. [53,54] has previously discussed the importance of allyl addition propene for the formation of olefins and aromatic from the thermal cracking of light hydrocarbons, and pyrolysis experiments by Simon and Back [23,24], Al-Shoaibi [9], and this study observed that C₅ and C₆ species started forming at very small extents of fuel conversion (cf. Fig. 4 and Tables S2–S5 in the SI).

The addition reactions of allyl radical to ethene and to 1,3-butadiene are also significant (the corresponding PESs are shown the SI; Figs. S4 and S5, respectively). Both of these PESs have recently been discussed as potential routes to MWG species formation [21]. The addition reaction to ethene is very similar to that of propene in that once the initial adduct is formed, it is energetically more favorable to cyclize than it is to dissociate back to reactants. The barrier for β -scission of cyclopentyl radical is comparable to the reverse ring-opening reaction and this leads to formation of H + cyclopentene. The addition reaction to 1,3-butadiene introduces another important feature for rapid MWG species formation. In this case, terminal addition leads to an adduct that is also resonantly-stabilized. The barrier for addition is lower than that for addition to ethene or propene and therefore the total rate constant is higher. The initial well is deeper, and the formation of the stabilized adduct is more favored. Several pathways that lead to stable cyclic species are also energetically avail-

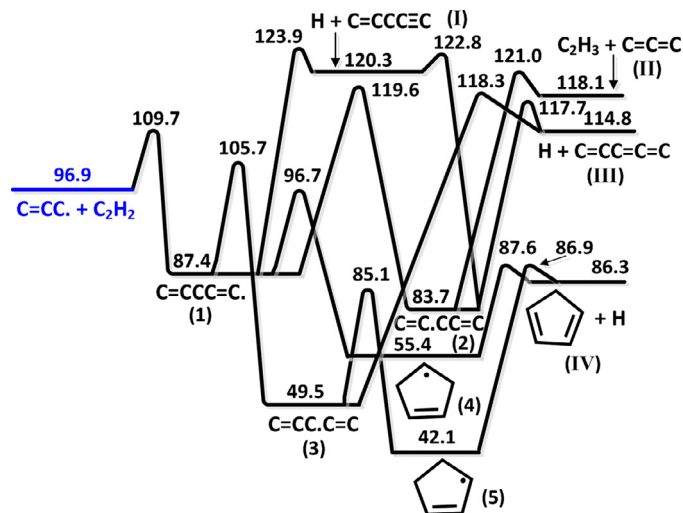
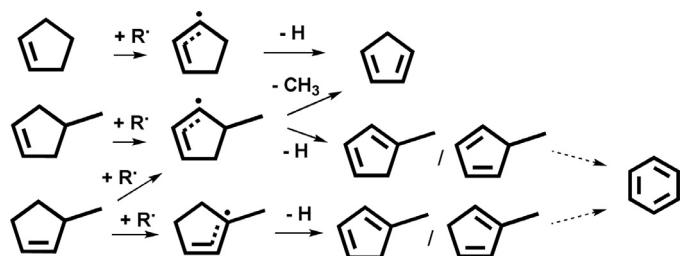


Fig. 11. Simplified C₅H₇ PES calculated at the CBS-QB3 level of theory, showing enthalpies in kcal/mol at 298 K. (Most H atoms in the species names are omitted for clarity.)

able. Allyl addition to 1,3-butadiene leads to the formation of CH₃ plus 1,4-cyclohexadiene and methyl-1,3-cyclopentadiene [21], both of which are important precursors for the benzene formation.

At sufficiently high temperatures the model predicts that significant amounts of acetylene may be produced. The addition reaction of allyl to acetylene is potentially important for the generation of cyclopentadiene, via the PES shown in Fig. 11. The initially formed adduct (1) can cyclize directly (4) or isomerize to its resonantly-stabilized isomer (3) that can cyclize (5) then form cyclopentadiene by β -scission of an H atom (IV). The energy barriers for either the isomerization or the β -scission reactions are well below the energy of the entrance channel. The PES can also be accessed by three other addition reactions (i.e., H atom addition to pent-4-en-1-yne (C≡CCC=C) (I) and 1,2,4-Pentatriene (C=CC=C=C) (III), or vinyl addition to allene (II)). These reactions would also lead to production of cyclopentadiene. The rate constant for the formation of cyclopentadiene + H from allyl addition to acetylene via this PES is consistent with the experimental measurement (shown in the SI, Fig. S6).

Another important resonantly-stabilized radical is cyclopentadienyl (cf. Fig. 9); reactions of this species need to be considered as potential pathways for MWG. For example, the addition reactions of cyclopentadienyl to cyclopentadiene were found to be important for formation of indene and styrene during the oxidation or thermal decomposition of cyclopentadiene [55]. We compared the addition reactions of cyclopentadienyl to ethene and propene to the analogous



Scheme 1. Subsequent reactions of the methyl-cyclopentene isomers and cyclopentene that lead to the formation of cyclopentadiene and benzene.

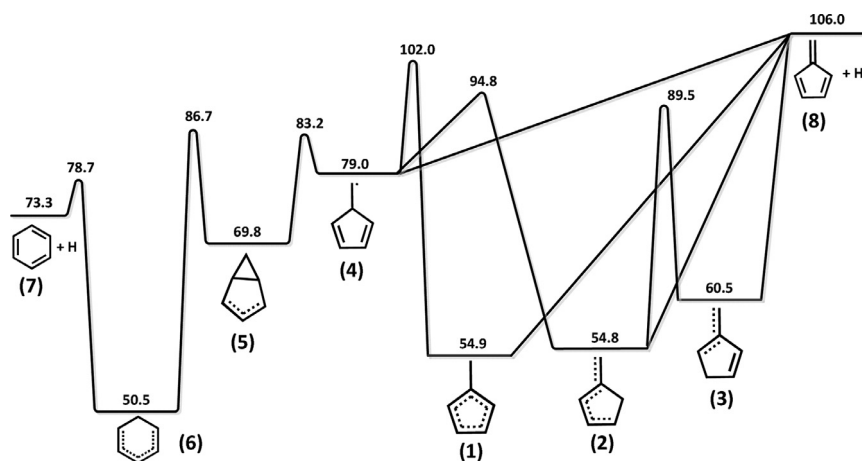


Fig. 12. Simplified C_6H_7 PES calculated at the CBS–QB3 level of theory, showing enthalpies in kcal/mol at 298 K.

reactions with allyl. The PES for cyclopentadienyl addition to ethene is included in the SI (Fig. S7). Although benzene and toluene are low energy exit pathways on this surface, the high barrier for isomerization of the initially-formed adduct makes these pathways relatively unimportant. The apparent rate constants for the reaction at 1 atm are shown in the SI (Fig. S8); the cyclic MWG product rate constants are much slower than those of allyl addition to ethene [21]. The PES for cyclopentadienyl to propene (including the terminal and non-terminal addition, constructed based on the rate estimation rules [21]) is shown in the SI Fig. S9, with the calculated apparent rate constants at 1 atm shown in the SI (Fig. S10). Compared to the rate constants for the reactions from allyl addition to propene (cf. Fig. 10), the rate constants for cyclopentadienyl addition to propene are lower by around one order of magnitude. This may be attributed to high barriers for the subsequent reactions (β -scission or H-atom shift via 3- and 4-member transition state rings) of the initially-formed adducts and therefore they are more likely to re-dissociate to the reactants. Thus, the addition reactions of cyclopentadienyl to the light olefin products appear to be much less important than the corresponding allyl reactions. Instead, cyclopentadienyl plays a more important role in forming larger MWG species (e.g., styrene, indene, and naphthalene) via recombination reactions, as discussed below.

The recombination between two resonantly-stabilized radicals provides another potential source for MWG formation. A number of experimental [56–60] and theoretical studies [61–63] have been performed for the self-recombination reaction of propargyl (C_3H_3) radicals and the reaction pathways leading to the formation of benzene have been well documented. The recombination between two allylic radicals (e.g., allyl + allyl, methyl-allyl + allyl, methyl-allyl + methyl-allyl, etc.) that lead to stable cyclic species after H-abstraction have also been investigated recently [21].

The recombination between methyl and cyclopentadienyl via the C_6H_7 PES forms methyl-2,4-cyclopentadiene. H-abstraction from the ring leads to the isomers (1), (2), and (3) in Fig. 12. These resonantly stabilized isomers are relatively quite stable and are unlikely to form benzene except at very high temperatures. On the other hand abstraction from the methyl carbon to form (4) would provide a more likely

route to benzene. However note that this PES does provide a route for conversion of fulvene to benzene via H-atom addition.

The self-recombination of two cyclopentadienyl radicals is important for the formation of naphthalene (after isomerization and loss of hydrogen), which has been studied by several groups [53–69]. Melius et al. [65] calculated the detailed PES, and proposed two sequential steps:



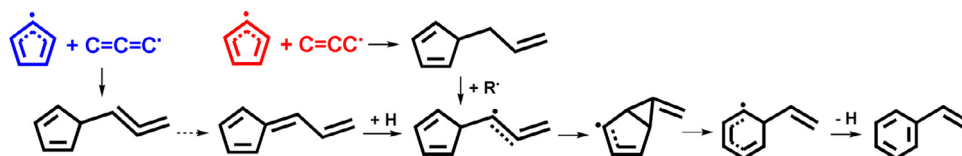
Unfortunately, rate constants were not reported in that study. There is substantial controversy regarding the treatment of this recombination reaction. Marinov et al. [66] proposed using one overall step to describe this process:



However, Lindstedt et al. [67] showed that this one-step reaction yielded naphthalene predictions that were inconsistent with measured trends in several fuels. Instead, they proposed a two-step sequence with the first reaction being weakly endothermic (8 kcal/mol).

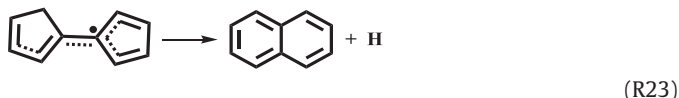


This treatment yields rate constants that over-predict the production of naphthalene observed in the current work. Recently this PES has been re-examined by several other theoretical studies [55,68,69]. However, the predicted results changed only marginally and the lack of information about the bimolecular reactions of the new intermediates added uncertainties as found by Nowakowska et al. [70] for the anisole oxidation. Furthermore, the calculated PES by Melius et al. [65] explicitly indicates many more reactions are involved. Therefore, in this study, instead of assigning rate constants for one or two over-



Scheme 2. Recombination between cyclopentadienyl and allyl, cyclopentadienyl and propargyl that lead to the formation of styrene.

all steps, we revisited the PES calculated by Melius et al., and assigned rate constants to each reaction step by analogy to similar recombination, H-atom transfer, cycloaddition, and H-atom addition reactions, as described in earlier publications [21,37,71,72]. This analysis indicated that the two most important reactions are:



The rate constants that are estimated in this study are much lower than those used in the earlier studies (cf. SI Table S7). When the full set of pressure-dependent rate constants was incorporated into the mechanism, the naphthalene production predictions captured the temperature and residence time dependence observed (cf. Fig. 4).

The recombination between allyl or propargyl with cyclopentadienyl can ultimately form styrene, after subsequent H-abstraction and isomerization (reaction Scheme 2). Sharma et al. [73] studied recombination between propargyl and cyclopentadienyl which also forms styrene. These are the two major pathways accounting for the formation of styrene in propene pyrolysis (cf. Fig. 4).

5. Predictions to other propene pyrolysis experiments

To test the robustness of the current model, simulations were performed for two sets of experiments where the initial conditions were substantially different for our experiments.

5.1. Highly diluted system

Davis et al. [27] reported a propene pyrolysis experiment using the Princeton Turbulent Flow Reactor where propene was highly diluted in N_2 at 1 atm (0.228 mol% propene); the pyrolysis temperature was 1210 K, with residence times ranging from ~10 to 160 ms. The measured major species were methane, acetylene, ethene, ethane, and propyne, and several MWG species were characterized, including 1,3-butadiene, 1-butene, and benzene. As shown in Fig. 13, when the residence time was shifted by 38 ms to match fuel conversion (to attempt to account for the uncertainty in fuel mixing zone), the product distributions were reasonably well predicted with the current model. The largest discrepancies occur for the formation of acetylene and allene. Acetylene is under-predicted at longer residence times. Allene is also under-predicted, although the time-dependence is similar to the experiment. Interestingly, the propyne predictions are in very good agreement with the experiment. Moreover, none of the other hydrocarbon products are over-predicted, so the cause of the discrepancy in acetylene and allene is not obvious. The model also predicts a significant amount of hydrogen, though the experiment was not designed to measure this product.

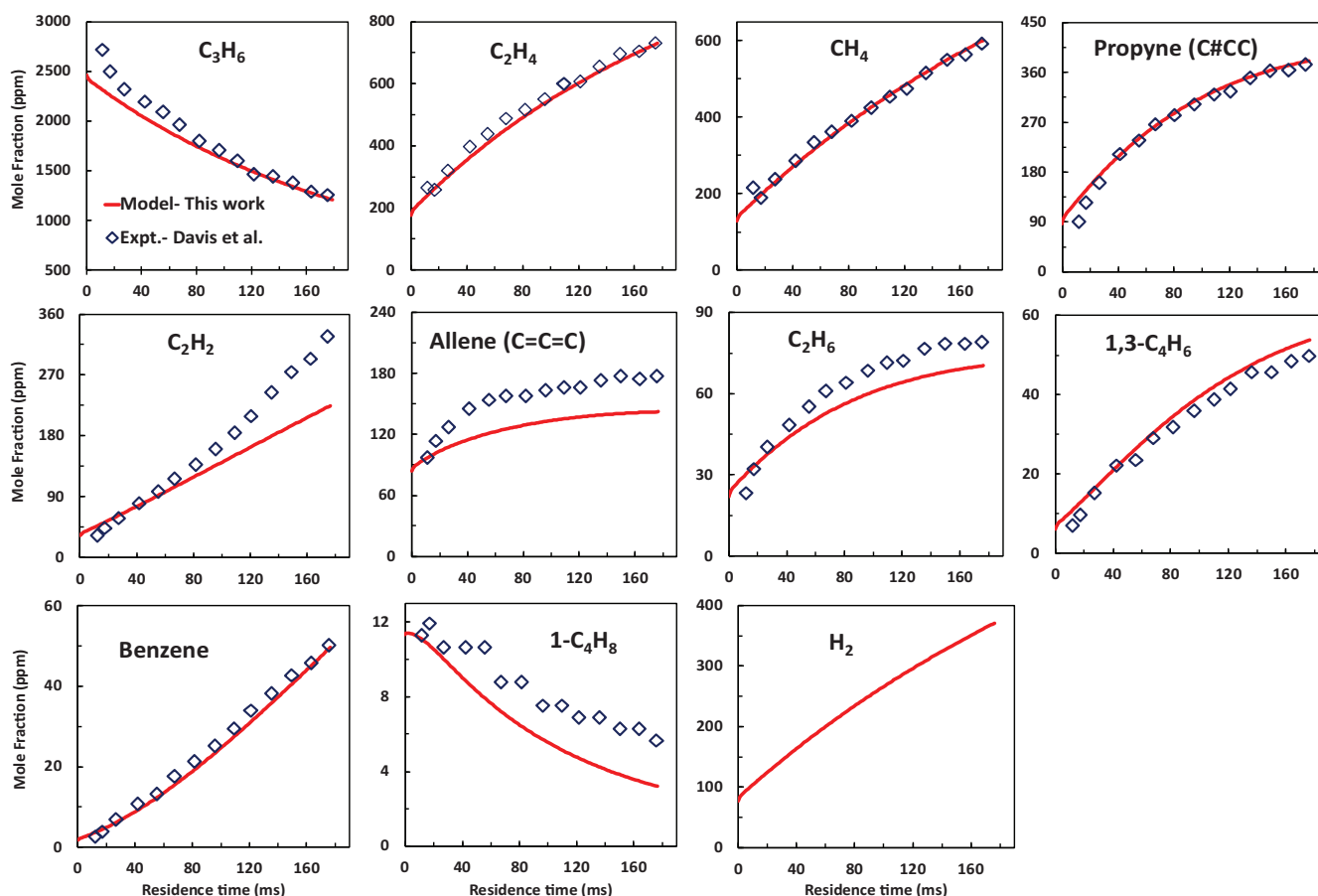


Fig. 13. Prediction of model to literature measurements of fuel conversion and formation of products with 0.288% propene diluted in N_2 at 1 atm, 1210 K (open symbols: experiment by Davis et al. [27]; lines: CSM model predictions with the residence time shifted by 38 ms to match the observed fuel conversion).

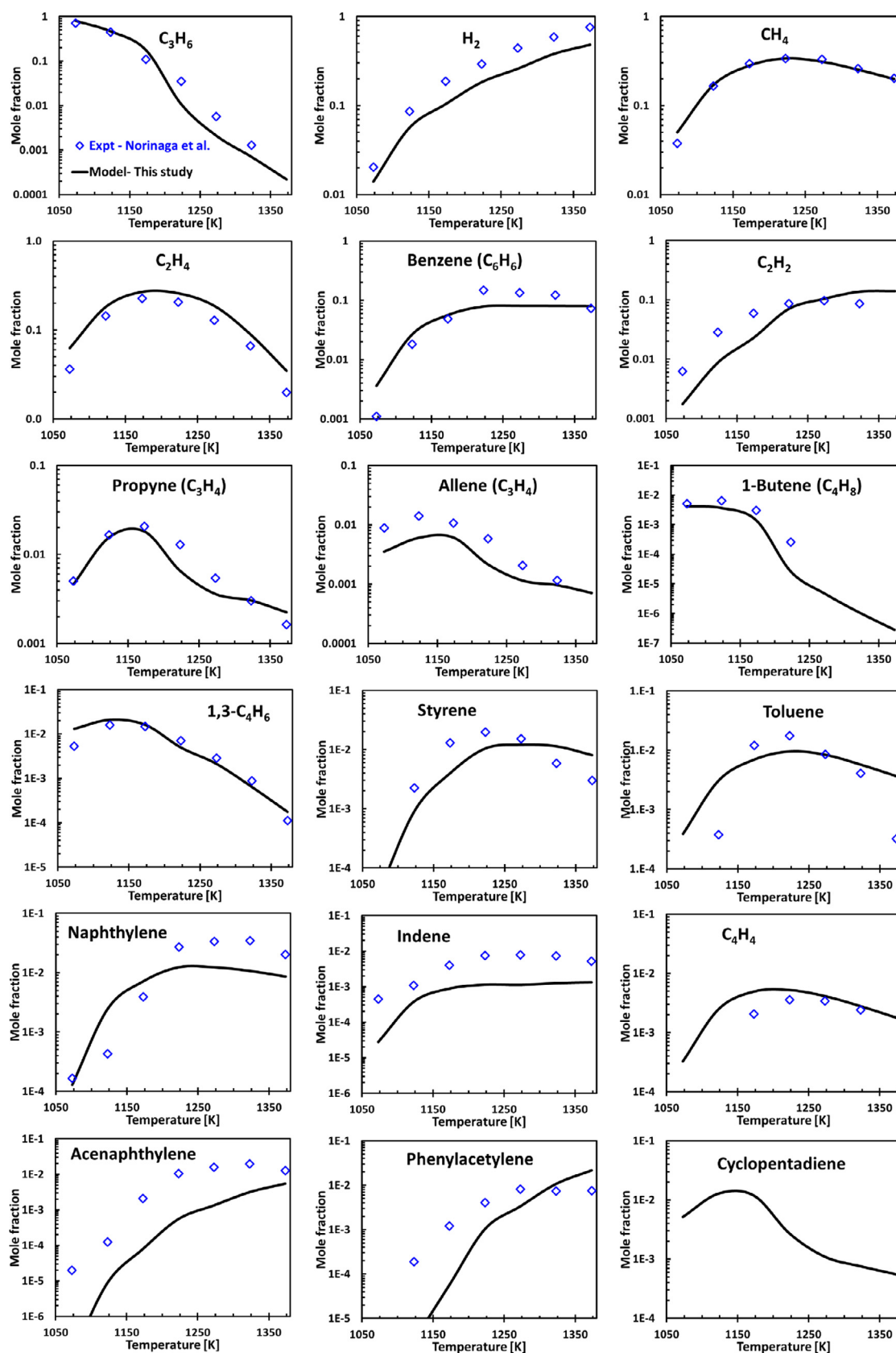


Fig. 14. Comparison of predicted mole fractions by the model of this study to a published experimental data for propene pyrolysis at $P = 8$ kPa and $\tau = \sim 0.5$ s as a function of temperature (symbols: experimental data by Norinaga et al. [29]; lines: CSM model predictions).

5.2. Neat fuel system

Norinaga et al. [29] conducted propene pyrolysis experiments without any dilution, at a low pressure of 8 kPa and temperature of 1075–1375 K with a residence time of 0.5 s. Fig. 14 compares the predictions using the model developed in this study to the experiment measurements. In general the results are very encouraging, especially considering the significantly higher temperature and the much lower pressure than in our experiments. The model captures the magnitude and the temperature dependence for the major species and many of the MWG species. Interestingly, these authors did not report the formation of any C₅ species, although the model predicts significant amount of cyclopentadiene.

6. Summary

The objective of this study is to characterize the kinetics of molecular weight growth (MWG) reactions that occur during propene pyrolysis. Propene was selected as a model fuel to validate an improved fundamentally-based kinetic model. It is the simplest olefin that can generate a resonantly-stabilized radical, and therefore it can serve as a model compound for larger olefins. Olefins are important initial products formed during the conversion of alkane fuels and therefore it is essential to accurately describe their thermal decomposition kinetics. A major improvement in this work, compared to several prior studies, was the inclusion of allyl addition reactions to propene. The model developed here captures the temperature and residence time impacts on propene conversion, major product production, and the production of MWG species. Sensitivity and rate of production analyses showed that the decomposition of propene is most likely initiated by the bimolecular reverse disproportionation reaction of propene, forming allyl plus *n*-iso-propyl under the conditions of this study. The propyl radicals can abstract from propene forming allyl and propane. The β -scission of the propyl radicals forms methyl and H atoms that can also abstract from propene, forming more “inert” allyl radicals. H atoms can add to propene, forming ethene and methyl. The subsequent allyl addition to propene plays a critical role in converting the allyl radical to more reactive methyl and H atoms to continue the chain, thus driving the propene decomposition. Including the allyl addition reaction to propene in several literature models, that were developed to describe alkane/alkene oxidation, substantially improved the predictions for the temperature dependence of propene conversion.

Multiple MWG species (C₄–C₁₁) were detected and quantitatively measured. The C₄ species start forming during the very early stages of propene conversion, followed by the formation of C₅ and C₆ products, and then C₇ products, while C₈–C₁₀ species are only generated after substantial conversion. The important pathways for the MWG species formation were analyzed. The reactions of resonantly-stabilized radicals were found to play an important role for the formation of MWG products. The recombination between two stabilized radicals or between a stabilized radical and an alkyl radical is important for the formation of noncyclic C₅ and C₆ products. The recombination adduct, after H-abstraction, can turn into cyclic species via subsequent cyclization and β -scission. The potential energy surface obtained from the addition of stabilized radical to olefins or diolefins shows that lower energy exit channels exist. Therefore, the formation of stable ring compounds via these surfaces is energetically favored. Given that resonantly-stabilized radicals are also likely to dominate in other olefin pyrolysis systems, the kinetic knowledge obtained here is expected to be applicable to these systems as well. Depending upon the conditions, improved understanding of olefin pyrolysis kinetics is also important for both alkane pyrolysis and oxidation models, especially for those models that are developed to describe MWG chemistry.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2015.08.012.

Table S1 lists the calculated residence times-temperature profiles from ChemKin. The experimental results are provided in Tables S2–S5 and the results of a sensitivity analysis are provided in Table S6. In Table S7 a comparison of the rate constants for the self-recombination reaction of cyclopentadienyl radical is provided. A diagram of the apparatus is provided in Fig. S1. Fig. S2 shows a comparison of the rate constants for the unimolecular reactions of propene. Fig. S3 shows a comparison of the rate constants for H-atom addition to propene as well as H-atom abstraction of propene by H-atom and CH₃ radical. Figs. S4 and S5 show the PESs for allyl addition to ethene and 1,3-butadiene, respectively. Fig. S6 shows a comparison of the rate constant for allyl addition to acetylene. The PESs and the apparent rate constants for the addition of cyclopentadienyl radical to ethene and propene are shown in Figs. S7/S8 and S9/S10, respectively. The kinetic mechanism and its associated thermodynamic data is provided in separate text files.

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